

The empirical recurrence for OEIS A250590, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A250590 records “Empirical recurrence of order 58” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 376$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 65$. Its last coefficient is $c_{58} = 77760 \neq 0$, so no further tail satisfies anything shorter; any order-58 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A250590 is “Number of $(n+1) \times (6+1)$ 0..2 arrays with nondecreasing $\max(x(i,j), x(i,j-1))$ in the i direction and nondecreasing $\min(x(i,j), x(i-1,j))$ in the j direction.”. It has offset 1 and begins

30952, 226985, 1347397, 7120483, 34503682, 157109231, 683598422, 2872989261, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:34:28, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 376$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 376$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 58: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 58.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 752$ exact terms from $n = 7$ on, it returns order 58 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	24	c_{16}	6260871402	c_{31}	−4822487174853	c_{46}	157354695665
c_2	−213	c_{17}	5251010631	c_{32}	−2418659845857	c_{47}	16816937603
c_3	543	c_{18}	−34449713322	c_{33}	7109741647650	c_{48}	−45091923492
c_4	4240	c_{19}	21532309435	c_{34}	−1804915697868	c_{49}	10958166135
c_5	−35542	c_{20}	100058099116	c_{35}	−5809933448410	c_{50}	5159357279
c_6	53031	c_{21}	−181706668596	c_{36}	4881370217918	c_{51}	−3296129604
c_7	449762	c_{22}	−111928101419	c_{37}	2033363834262	c_{52}	208388484
c_8	−2172831	c_{23}	612867999093	c_{38}	−4649570226104	c_{53}	322785792
c_9	141207	c_{24}	−298135681074	c_{39}	970111965049	c_{54}	−99878904
c_{10}	23694983	c_{25}	−1147330109322	c_{40}	2357801258909	c_{55}	−449280
c_{11}	−53734866	c_{26}	1620872475646	c_{41}	−1622987102261	c_{56}	5709744
c_{12}	−92752480	c_{27}	891733649745	c_{42}	−453891085065	c_{57}	−1157328
c_{13}	576489563	c_{28}	−3589166037197	c_{43}	920491855045	c_{58}	77760
c_{14}	−368052755	c_{29}	1253550350732	c_{44}	−192065651697		
c_{15}	−2872266894	c_{30}	4469322100422	c_{45}	−256651499635		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 65$, and it is the recurrence OEIS A250590 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{58} - c_1 x^{57} - \dots - c_{58}$. Its constant term is $-c_{58} = -77760$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 58 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 58, so $\deg q = 58$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 58, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 58, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 77760.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-58 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A250590.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.